

GRADUATE TEACHING · TWO × 90 MINUTES

Fairness and Bias in AI

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Learning Objectives

01

Define the triad

Name bias 偏差, fairness 公平, and harm 损害 without collapsing them.

02

Distinguish kinds

Separate group vs individual, allocative vs representational fairness.

03

Compute metrics

Calculate DP, equal opportunity, equalized odds, PPV, calibration.

04

Locate bias

Trace error through data, labels, models, deployment, and loops.

05

Compare remedies

Contrast pre-/in-/post-processing, HITL, monitoring, and audit.

06

Plan an audit

State impossibility results; draft a generative / legal audit plan.

Course Roadmap

Session 1 · 90 min

OPENING

Foundations

Terms, harms, classic cases, then news and GitHub files.

AUDIT

Audit and fix

Datasheets, dashboards, pre-/in-/post-processing, HITL.

Session 2 · 90 min

SOURCES

Sources of bias

Measurement, sampling, labels, proxies, hiring, credit.

GENAI

Generative systems

LLM bias, RLHF, multimodal, red-team, incidents.

METRICS

Metrics

Group criteria, individual and causal fairness, impossibility.

GOVERN

Governance

NIST, EU, PRC, Hong Kong; capstone audit and rubric.

Why Fairness Matters

Allocative harm 分配性损害

Opportunity, money, or liberty is withheld or granted by a score.

Hiring ranks, credit limits, pretrial detention, benefit eligibility.

Error has an immediate, countable cost to a person.

Classic test: who is denied, and at what false-positive rate?

Representational harm 表征损害

A group is stereotyped, erased, or mocked in outputs. Image models, search rankings, generated biographies.

Harm is cultural and cumulative, not a single denial.

Classic test: who is depicted, and who never appears?

Crawford, The Trouble with Bias, NIPS keynote 2017; Barocas, Hardt, and Narayanan, Fairness and Machine Learning, 2023.

Bias, Fairness, and Harm

Bias 偏差

A systematic deviation from a stated target.

Statistical: estimator vs parameter.

Sociotechnical: a process that tilts outcomes by group.

Bias is a property of a pipeline, not a person.

Fairness 公平

A family of criteria, not one number.

Group: equal rates across A.

Individual: similar people, similar treatment.

Always name the metric and the group.

Harm 损害

The lived consequence of a decision.

Denial, stigma, surveillance, erasure.

Fairness metrics are proxies for harm.

Ask: who is worse off if we ship this?

Cases That Changed the Field

UNITED STATES · 2016

COMPAS

Broward County, FL; $n \approx 7,214$ with 2-year follow-up.

Black defendants: 45% FPR vs 23% for White.

White reoffenders: 48% FNR vs 28% for Black.

Overall accuracy similar (~61%).

Angwin et al., ProPublica, 23 May 2016.

UNITED STATES · 2018

Amazon hiring

Resume models trained on 10 years of mostly male applicants.

Penalized “women’s” and two women’s colleges.

Edited terms; residual proxies remained.

Project disbanded; not sole hiring tool.

Dastin, Reuters, 10 Oct 2018.

FAT* · 2018

Gender Shades

3 commercial gender classifiers.

Darker-skinned females: error up to 34.7%.

Lighter-skinned males: error $\leq 0.8\%$.

IJB-A 79.6% lighter; Adience 86.2%.

Buolamwini and Gebru, PMLR 81, 2018.

More Cases: News and GitHub

Five extra files. Open-domain news plus public GitHub repos. Same rule: date, jurisdiction, source.

HEALTH

Cost proxy

Obermeyer 2019.
Science.

STATE

Exams, welfare

Ofqual 2020. Dutch
benefits.

GITHUB

PRs, Copilot

Terrell 2017. Pearce
2022.

PLATFORM

Image crop

Twitter research
repo. BBC 2021.

HIRING

Screens, law

iTutorGroup. NYC
LL144.

Healthcare: Cost as a Proxy

Obermeyer et al., Science 366(6464):447-453, 25 Oct 2019. US commercial care-management algorithm.

What it did

Predicted cost, not illness. Millions of patients. At the same risk score, Black patients were sicker.

What followed

Black patients in extra-care programs: 17.7%. Relabeling need would raise that to 46.5%.

Classroom cut

A high-AUC proxy can still be a construct error. Spend is not need. This is measurement bias, not a bad ROC.

Public Sector: Exams and Benefits

ENGLAND · AUG 2020

Ofqual A-level algorithm

Exams cancelled. Model used teacher ranks plus school history.

About 39% of A-level results downgraded (CNBC, 21 Aug 2020).

State-school high-achievers hit harder. Grades withdrawn; teacher predictions restored.

NETHERLANDS · FROM 2013

Childcare-benefit risk score

Tax authority flagged claims as possible fraud. Non-Dutch nationality raised the score.

Tens of thousands of families wrongly accused (Amnesty, 25 Oct 2021).

Cabinet fell Jan 2021. Black-box review with no reason codes.

GitHub: PRs and Copilot

OPEN SOURCE · PEERJ 2017

Pull-request gender study

Terrell, Kofink, Middleton et al., PeerJ Computer Science 2017.

Women: 78.7% PRs accepted vs 74.6% for men (NCSU, 1 May 2017).

Outsiders with identifiable gender: women 62.5% vs 71.8% when profiles were gender-neutral (BBC quoting the paper, 12 Feb 2016).

COPILLOT · IEEE S&P 2022

Asleep at the Keyboard?

Pearce, Ahmad, Tan, Dolan-Gavitt, Karri. IEEE S&P 2022. arXiv:2108.09293.

89 CWE-style scenarios; 1,689 programs.

About 40% of Copilot completions were vulnerable.

Training on public GitHub copies insecure patterns.

Twitter Crop: Open Repo

User finding

Sep 2020: previews preferred a White face over a Black face regardless of order (BBC, 20 May 2021).

Same pattern in paired photos, including public figures.

Company numbers

Saliency model, 2018, trained on eye-tracking.

Vs 50-50: +8% women over men; +4% White over Black; +7% White women over Black women (BBC 2021).

They later removed auto-crop.

GitHub artifact

github.com/twitter-research/image-crop-analysis

Yee, Tantipongpipat, Mishra. PACM HCI / CSCW 2021.

arXiv:2105.08667. Apache-2.0 notebooks.

Lesson: some tasks should not be an algorithm.

Hiring Software and NYC LL144

EEOC · 2022-2023

iTutorGroup screen

Software auto-rejected women 55+ and men 60+. Over 200 applicants. Consent decree: USD 365,000 (EDNY, 2023).

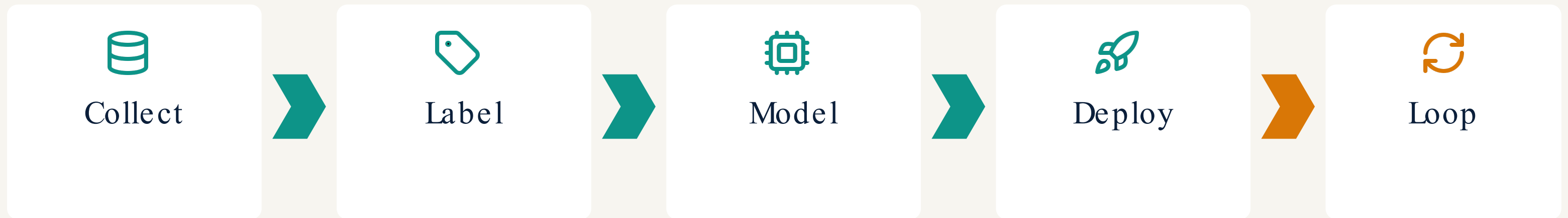
EEOC tied the case to its AI and algorithmic fairness initiative. Akin Gump (21 Sep 2023): may be a programmed rule, not generative AI. Still an automated employment decision.

NEW YORK CITY

Local Law 144

AEDT may not be used unless a bias audit ran within one year, a summary is public, and notice is given.
DCWP enforcement from 5 Jul 2023 (nyc.gov/dca).
Classroom: audit, publish, notify. Not a US federal AI Act.

Data Lifecycle of Bias



Where bias can enter —treat each stage as an audit checkpoint

Collect: who is sampled, measured, consented.

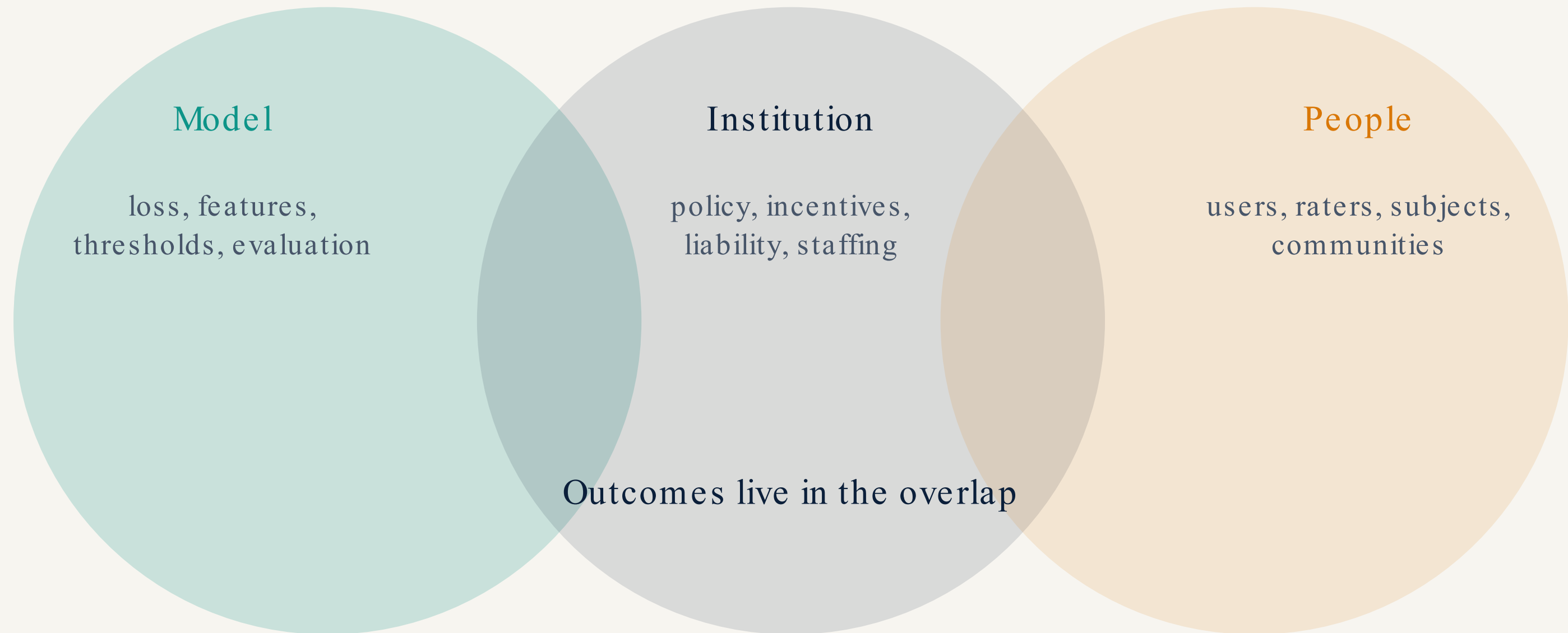
Label: whose judgments become ground truth.

Model: objective, features, capacity, evaluation slice.

Deploy: threshold, override, affected population.

Loop: logged outcomes become tomorrow's training data.

A Sociotechnical Frame



Selbst et al., Fairness and Abstraction in Sociotechnical Systems, FAT* 2019. Fairness is not a model-only property.

Sensitive Attributes and Proxies

Kind	Examples	Why it matters
Protected 受保护属性	Race, sex, disability, religion, age	Named in anti-discrimination law; lists are jurisdiction-specific
Direct use	Feature A inside the model	Often restricted; still needed later to measure gaps
Proxy 代理变量	ZIP, name, language, browsing	Correlated with A; dropping A does not drop the signal
Unawareness	“We did not use race”	Fails when proxies remain; also blocks disparity audits

Legal lists differ: US Title VII / ECOA; EU Charter Art. 21; HK Sex Discrimination Ordinance. Always name the jurisdiction. [VERIFY] the local protected-class list for any deployment.

Checkpoint: Foundations Quiz

Q1

Is “the model is 90% accurate” a fairness claim? Why or why not?

Q2

Name one allocative and one representational harm from S04.

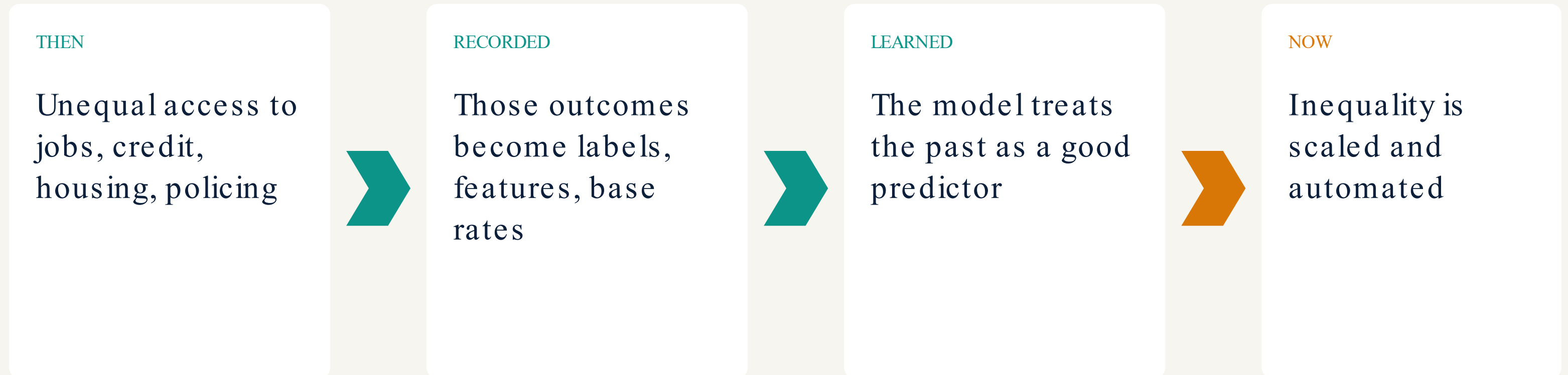
Q3

Which lifecycle stage is Amazon’s “women’s chess club” penalty?

Q4

Why does dropping the protected attribute often fail?

Historical and Structural Inequality

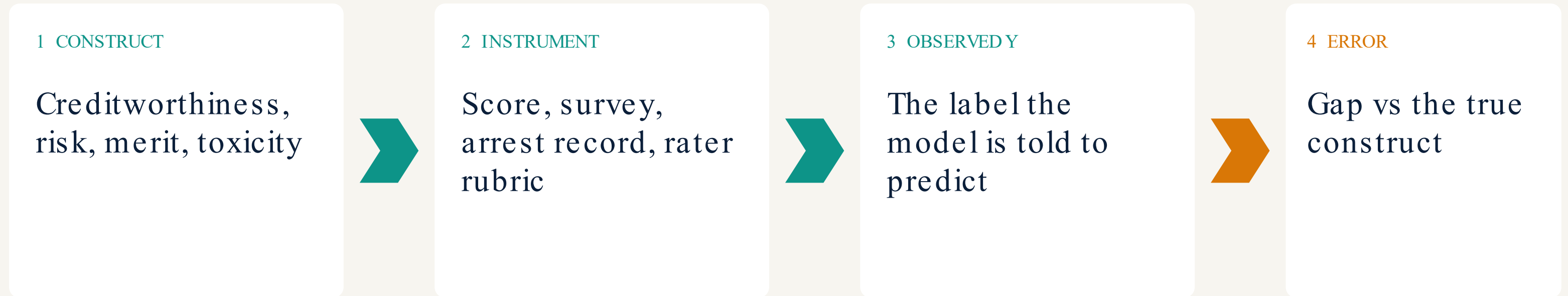


A historically accurate predictor can still be an unfair policy.

Accuracy on yesterday's world is not a moral defense of tomorrow's allocation.

Barocas and Selbst, Big Data's Disparate Impact, 104 Calif. L. Rev. 671 (2016).

Measurement Bias



Classic mismatch

Arrest $\neq$ offense. Health spend $\neq$ health need. Clicks $\neq$ relevance.
If the instrument is biased, a “fair” model of Y still harms.
Jacobs and Wallach, Measurement and Fairness, FAccT 2021.

Representation and Sampling

Who is in the data?

Coverage: some groups never appear.

Density: some groups appear only as outliers.

Quality: photos, transcripts, labels differ by group.

Gender Shades: IJB-A 79.6% lighter-skinned; Adience 86.2%.

Result: decision boundaries fit the majority mode.

Audit questions

What is the sampling frame?

Which languages, devices, regions are missing?

Are rare intersections below a power threshold?

Was consent obtained for every source?

Would adding data change the metric, or the construct?

Labels and Annotation

Failure	What it looks like	Who is hurt	Check
Disagreement	Raters split on the same item	Contested identities, dialects	Report κ / raw %
Rater pool	One demographic labels all	Out-groups in the corpus	Rater demographics
Noisy gold	“Ground truth” is a majority vote	Minority interpretations	Keep disagreement

Toxicity, hate, and “professional” labels are value judgments, not sensors.
 Sap et al., The Risk of Racial Bias in Hate Speech Detection, ACL 2019.
 Design choice: preserve disagreement instead of forcing a single gold label.

Missingness

MCAR

Missing completely at random.
Rare in socio-technical data.
Dropping rows is least harmful here.

MAR

Missing depends on observed X.
Imputation can work if the model of missingness is right.
Still check group-wise rates.

MNAR

Missing depends on the unobserved value itself.
Income, diagnosis, arrests: non-response is the signal.
Treat as a fairness failure, not a stats nuisance.

Little and Rubin, Statistical Analysis with Missing Data. In credit and health, missingness often tracks protected class. Always report missing rates by group.

Proxy Variables

ZIP / district

Residential segregation encodes race and class.

Redlining history is in the geography, not the model card.

Ask: would a different district flip the score?

Name

Names correlate with gender, ethnicity, language.

Resume screening and search rank leak this signal.

Amazon: the word “women’s” was an explicit proxy.

Language

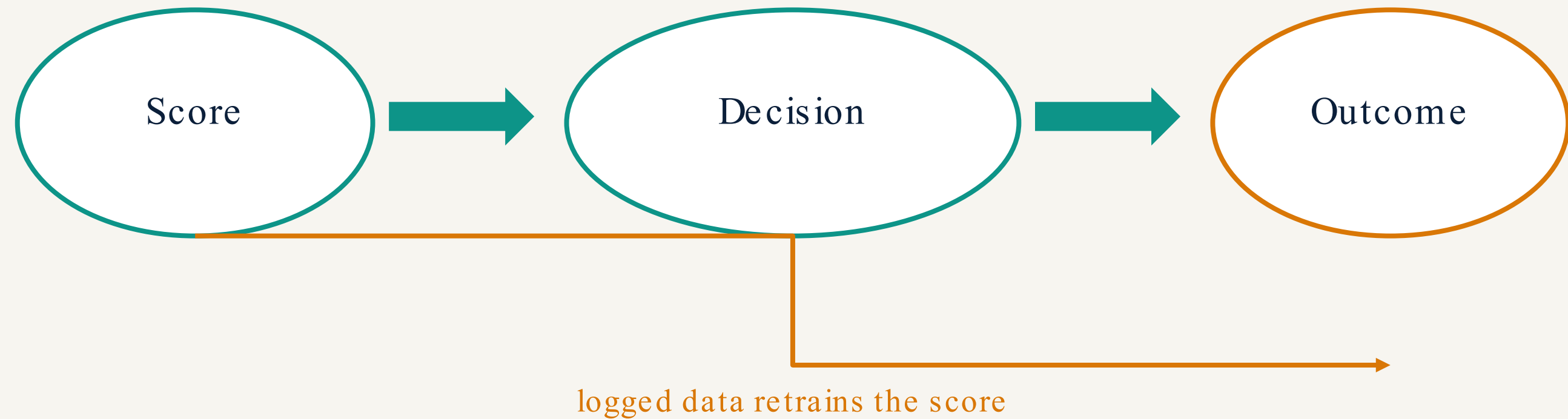
Dialect, code-switch, and CJK vs Latin script.

Toxicity and ASR error rates track language variety.

A “neutral English” filter is not a fairness control.

Prince and Schwarcz, Proxy Discrimination in the Age of Artificial Intelligence, 2020. Test: can a second model predict A from X?

Feedback Loops



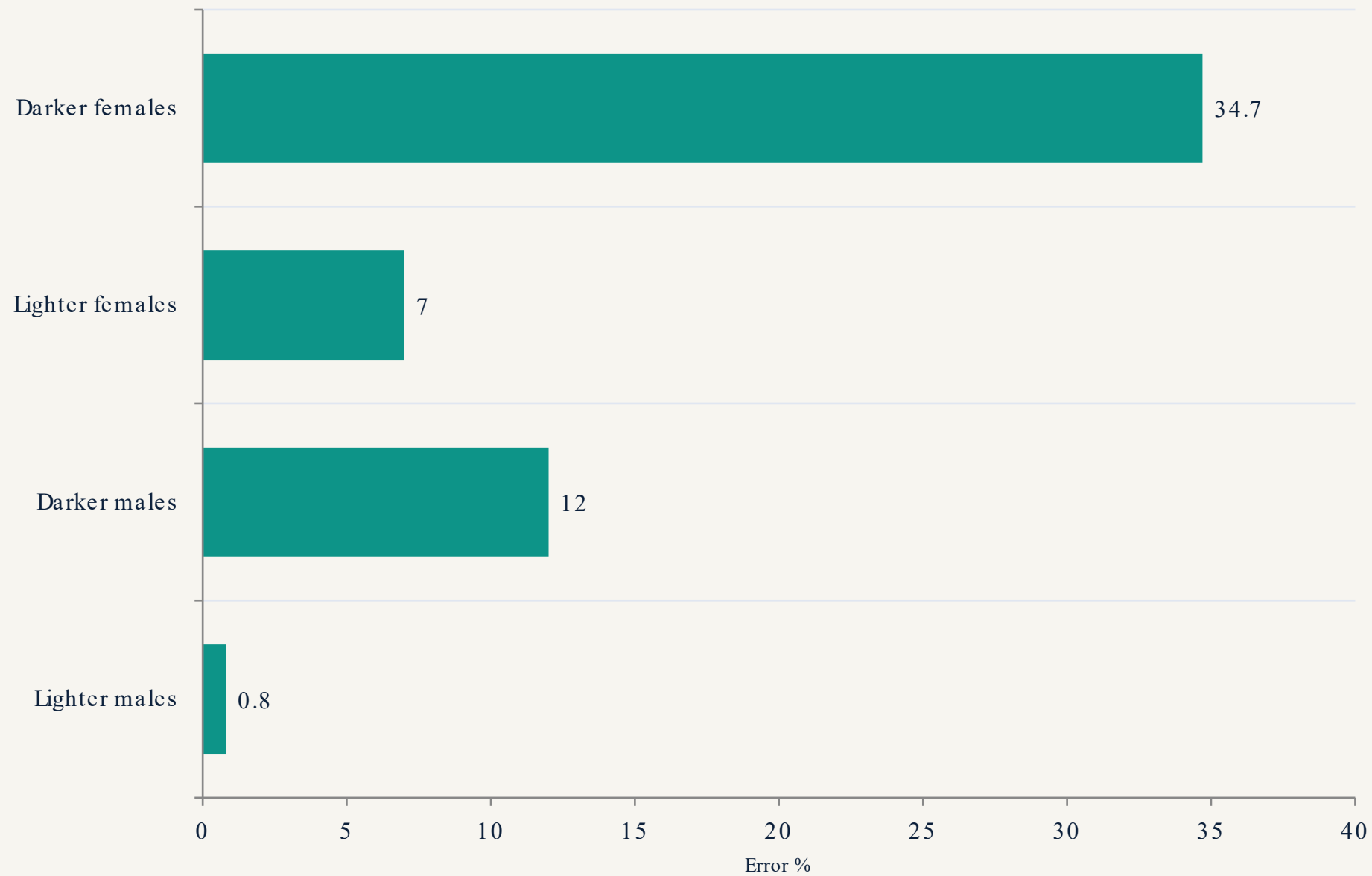
Policing: more patrols → more arrests → higher predicted risk → more patrols.

Credit: denial → thinner file → more denial. Ranking: clicks → more rank.

Ensign et al., Runaway Feedback Loops in Predictive Policing, FAT* 2018.

Intersectionality

Gender classification error rates, Gender Shades (Buolamwini and Gebru, 2018).



Single-axis hides the worst cell

Average-by-gender looks fine.
 Average-by-skin-type looks better than the intersection.
 Crenshaw, 1989. Always report the Cartesian product, then small-n caveats.
 Lighter males $\leq 0.8\%$. Darker females up to 34.7%.

Hiring Case: Amazon

Dastin, Reuters, 10 Oct 2018. Experimental resume ranker, 2014–2017. Not a sole hiring tool.

DATA

10 years of submitted resumes.
Most came from men in tech.
~500 models; ~50,000 terms.

FAILURE

Penalized “women’s ...”.
Downgraded two women’s colleges.
Favored verbs common on men’s CVs.
Editing terms did not kill proxies.

LESSON

Historical hiring is not a target.
Word filters are not an audit.
Need a construct of “job skill” independent of past gender mix.

Credit Case: Risk Scoring

What is scored

Probability of default or late payment.

Features: payment history, utilization, inquiries, thin-file markers.

Outcome: price, limit, or denial.

Where bias enters

Historical access to credit is unequal.

ZIP and shopping data act as proxies.

Label = observed default, which depends on who was granted credit.

What to audit

Approval rate by group (DP).

TPR among true good risks (EO).

Calibration of $p(\text{default})$.

US ECOA / Reg B; HK PDPO if personal data. No invented Apple Card stats here.

Classification Evaluation Review

Symbol	Name	Meaning
TPR / Recall	True positive rate	$P(\hat{Y}=1 \mid Y=1)$ among people who should receive the good
FPR	False positive rate	$P(\hat{Y}=1 \mid Y=0)$ among people who should not
PPV	Precision / predictive value	$P(Y=1 \mid \hat{Y}=1)$ among those flagged
Base rate	Prevalence	$P(Y=1 \mid A=a)$ —often unequal across groups

Accuracy mixes these rates with the base rate. Two groups can share accuracy and still have very different TPR and FPR. That is the COMPAS debate in one sentence.

Confusion Matrix

	$\hat{Y}=1$ predicted +	$\hat{Y}=0$ predicted −
$Y=1$	TP	FN
$Y=0$	FP	TN
Read as	$TPR = TP/(TP+FN)$	$FPR = FP/(FP+TN)$

Build one matrix per group $A=a$. Fairness metrics are constraints across those matrices. $PPV = TP/(TP+FP)$. Never compare a single mixed matrix and call it fair.

Demographic Parity

$$P(\hat{Y} = 1 \mid A = a) = P(\hat{Y} = 1 \mid A = b) \quad \text{independence of } \hat{Y} \text{ and } A$$

When people use it

Advertising, outreach, candidate slates.

Y is unobserved or contested.

Easy to explain to non-statisticians.

Failure modes

Ignores who is qualified (Y).

Can require accepting unqualified or rejecting qualified people if base rates differ.

Illegal in some US credit settings as a quota.

Equal Opportunity

$$P(\hat{Y} = 1 \mid Y = 1, A = a) = P(\hat{Y} = 1 \mid Y = 1, A = b) \quad \text{equal TPR}$$

Hardt, Price, and Srebro, Equality of Opportunity in Supervised Learning, NeurIPS 2016

Only people who truly deserve the favorable label ($Y=1$) are compared.

False negatives are equalized; false positives are not constrained.

Natural for lending, hiring, and benefits: do not miss qualified members of any group.

Still needs a defensible Y . If Y is biased, equal TPR equalizes a bad construct.

Equalized Odds

Equal TPR and equal FPR across A. $\hat{Y} \perp A \mid Y$

What it asks

Qualified people are found at the same rate.

Unqualified people are flagged at the same rate.

COMPAS: ProPublica's FPR gap is an equalized-odds failure.

Cost

Stricter than equal opportunity.

Usually requires group-specific thresholds.

Incompatible with calibration when base rates differ (next slides).

Predictive Parity and Calibration

Predictive parity

$P(Y=1 \mid \hat{Y}=1, A=a)$ equal. Same PPV.

A “high-risk” flag means the same chance of the outcome.

Northpointe’s reply to ProPublica used this family of tests.

Dietrich, Mendoza, and Brennan, Northpointe, 2016.

Calibration

$P(Y=1 \mid S=s, A=a) = s$ for every score bin s .

A 70% score is a 70% event in every group.

Stronger than PPV of a single threshold.

Needed for pricing, ranking, and human trust in the number.

Individual Fairness

$$D(M(x), M(x')) \leq d(x, x') \quad \text{similar people, similar treatment}$$

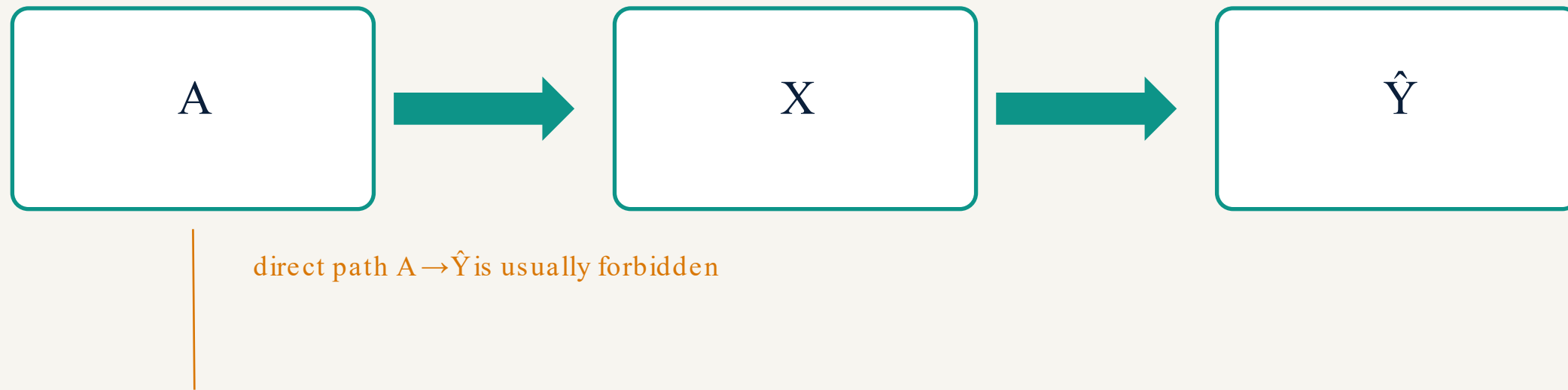
Promise

Protects a person, not only a group average.
Catches arbitrary flips between near-neighbors.
Dwork et al., Fairness Through Awareness, ITCS 2012.

Hard part

You must define the task metric d .
If d ignores a protected attribute that should not matter,
you smuggle bias in.
Group metrics can still fail even if Lipschitz holds on a
bad d .

Causal Fairness



Counterfactual fairness: $\hat{Y}(A \leftarrow a) = \hat{Y}(A \leftarrow a')$ for the same person (Kusner et al., 2017).

Path-specific: some mediators (e.g. hours worked) may be allowed; others (ZIP) not.

Requires a causal graph. If the graph is wrong, the fairness claim is wrong.

Ranking and Exposure

Not a binary $\hat{Y}$

Search, recommender, RAG, hiring slates.

Position 1 gets most clicks; position 20 is invisible.

Fairness = share of exposure, not a single accept/reject.

Singh and Joachims, Fairness of Exposure, SIGIR 2018.

Audit skeleton

Define items and groups (authors, vendors, dialects).

Measure discounted exposure by rank.

Compare to relevance or to equal opportunity in ranking.

Watch popularity feedback: exposure today becomes features tomorrow.

Impossibility and Trade-offs

Kleinberg, Mullainathan, and Raghavan, ITCS 2017 (arXiv:1609.05807). Except in special cases, these three cannot hold together.

Calibration

Score $s \approx P(Y=1 \mid s)$ in each group.

Balance for +

Equal TPR: people with $Y=1$ have the same expected score.

Balance for −

Equal FPR: people with $Y=0$ have the same expected score.

Special cases: equal base rates, or a perfect predictor. Otherwise choose. Chouldechova 2017 is the PPV vs equalized-odds twin result. Write the choice down; do not hide it in a default threshold.

Data Audit

1

Provenance

Who collected, when, under what license.
Web scrape vs consented panel vs admin records.

2

Gaps

Missing groups, languages, devices.
Missingness rates by A. Label disagreement.

3

Intended use

What the data must not be used for.
Gebru et al., Datasheets for Datasets, CACM 2021.

Subgroup Metrics

Slice	Why	Watch-out
Single attribute	Legal categories, simple dashboard	Hides intersections
Cartesian product	Worst-off cell, Gender Shades lesson	Cell n collapses; CIs explode
Error clustering	Find unknown slices (multicalibration)	Post-hoc groups need a story
Small-n rule	Suppress or widen interval if $n < k$	Silence can hide harm; document k

Hébert-Johnson et al., Multicalibration, COLT 2018. Report n per cell next to every rate. Do not color a tiny cell “green” because the point estimate looks good.

Uncertainty and Intervals

Do not ship a point

A 4-point TPR gap may be noise.

Binomial / bootstrap CIs on each group rate.

Then a CI on the difference, not two separate stars.

Multiple slices inflate false “gaps.” Pre-register primary A.

Classroom rule

Every fairness number on a dashboard has n , CI, and date.

If the interval covers zero, say “inconclusive,” not “fair.”

If n is tiny, collect more data or widen the group —do not invent a story.

Fairness Dashboard



Metrics

TPR, FPR, PPV,
calibration by group.
Named criteria, not a
single fairness score.



Slices

Primary A, intersections,
time. n and CI on every
tile.



Drift

Population mix, base
rates, feature shift. An
alert is not an auto-fix.



Owner

Named human, SLA,
rollback path. A chart
without an owner is
decoration.

Model Cards and Datasheets

Datasheets for Datasets

Geburu et al., CACM 64(12), 2021.

Motivation, composition, collection, preprocessing, uses, distribution, maintenance.

Answers “what is in this table?” before any model is fit.

Model cards

Mitchell et al., Model Cards for Model Reporting, FAT* 2019.

Intended use, metrics by group, evaluation data, ethical considerations, caveats.

A card is not a certificate. It is a structured confession of scope.

Pre-processing

Reweight

Up-weight underrepresented (x, y, A) cells.
Kamiran and Calders, 2012.
Leaves the model class unchanged.

Resample / repair

Oversample, undersample, or edit features toward independence (Feldman et al.).
Can distort constructs. Check utility.

Limit

Cannot fix a biased label Y.
Downstream trainers may undo the repair.
Document the transform in the datasheet.

In-processing

Constrained loss

Add a fairness penalty or constraint to training.

Zafar et al.; Agarwal et al. reductions.

Need A at train time.

Adversarial

Train a predictor that fools a group discriminator (Zhang, Lemoine, Mitchell).

Unstable; check residual proxies.

Limit

Optimizes the metric you wrote, not harm.

May trade utility sharply.

Re-evaluate after every checkpoint, not once.

Post-processing

Keep the model. Change the decision rule $\hat{Y} = 1 \{S \geq t_A\}$.

Hardt et al. 2016

Derive group thresholds for equalized odds from ROC curves.

No retraining. Needs A at decision time.

Transparent: the policy is the threshold table.

Watch-outs

Using A at decision time can be legally restricted.
Does not fix a biased S —it only rebalances errors.
Randomized thresholds are statistically clean and politically hard.

Threshold Trade-offs

One t for all

Simple, same rule.
Usually unequal TPR/FPR if base rates or score distributions differ.

Group t_A

Can hit equal opportunity or odds.
Looks like different bars for different people —legal and ethical review required.

Who chooses?

Not the default 0.5.
Not the F1 maximizer alone.
Name costs of FP vs FN, then pick t.
Record it in the model card.

Human-in-the-Loop



Humans can correct the model—or rubber-stamp it under time pressure (automation bias).

Design: when review is mandatory, what the reviewer sees, how override is recorded, who is reviewed more often.

Green and Chen, The Principles and Limits of Algorithm-in-the-Loop Decision Making, CSCW 2019.

Generative AI Bias

Stereotype

Occupations, dialects, and regions default to a majority prototype.
Representational harm in text and images.

Omission

Languages, legal systems, and histories never appear.
HKGAI-V1 (Han et al.,
arXiv:2507.11502, preprint) targets
HK multilingual and legal context for
this reason.

Allocation

Generated advice, rankings, and code can still deny opportunity.
Audit both the prompt distribution
and the output distribution.

LLM Preference and RLHF

Whose preference becomes the reward? Helpfulness $\neq$ harmlessness.

PKU-SafeRLHF

Ji et al. (incl. Sirui Han, Yike Guo), ACL 2025.
44.6k prompts; 265k QA pairs; 19 harm categories; 3 severity levels.
166.8k preference pairs; dual helpfulness / harmlessness labels.

Classroom takeaway

A single scalar reward collapses a trade-off.
Rater pool, instructions, and severity labels are fairness choices.
Do not cite this as a Han solo paper.

Multimodal and Vision Bias

Vision classification

Gender Shades: darker-skinned females up to 34.7% error vs $\leq 0.8\%$ for lighter-skinned males.

Dataset composition (IJB-A, Adience) predicted the gap. Intersectional slices first, then averages.

Safe RLHF-V

Ji et al. (incl. Sirui Han), arXiv:2503.17682, revised May 2025 (preprint / NeurIPS 2025 venue listing).

BeaverTails-V dual preference; Beaver-Guard-V.

Paper reports +34.2% safety and +34.3% helpfulness under their metrics —cite as authors' evaluation, not an independent audit.

Retrieval and Ranking Bias



The generator can only be as fair as the retrieved set.

Audit index coverage (language, jurisdiction, publisher) and exposure by rank.

HKGAI-V1 uses RAG for timely, locally grounded answers —retrieval policy is a fairness control, not a detail.

Fairness, Safety, Privacy

Fairness

Needs group labels to measure gaps.

Collecting A can itself be sensitive.

Safety

Refusal and filters can silence minority dialects or legal topics.

SafeLawBench (Cao et al. incl. Han, ACL Findings 2025): legal-standard safety, 24,860 MCQs; SOTA < 80.5%.

Privacy

PrivaCI-Bench (Li et al. incl. Han, ACL 2025): privacy as contextual integrity, not only PII.

LLMs capture CI parameters yet still fail compliance.

Limits of Debiasing

Word lists fail

Amazon neutralized “women’s” and still expected residual sorting (Reuters 2018).

The model finds another correlate.

Embedding surgery

Gonen and Goldberg, Lipstick on a Pig, 2019: gender direction removal is incomplete.

Geometry can look fair while clustering remains.

What remains

Change the construct, the data, the decision, or the product scope.

Debiasing is a control, not a certificate.

Red-teaming

Stereotype probes

Occupations, national values, dialects.

NaVAB (Shi et al. incl. Han, arXiv:2504.12911, preprint): CN, US, UK, FR, DE value alignment.

Legal-safety probes

SafeLawBench three-tier legal risks; refusal vs inaccurate advice.

PrivaCI-Bench for inappropriate information flows.

Process

Written threat model. Diverse attackers. Log successes. Fix, then retest the same corpus.

Red-team is sampling, not proof of safety.

Deployment Monitoring

Signal	Example	Action
Population mix	New dialect, age band, region	Recompute slices; may need new labels
Performance	TPR drop in one group	Page owner; hold rollout
Feedback	Appeals, complaints, news	Treat as labeled incidents
Model change	Prompt, RAG, weights, threshold	Re-run the fairness suite before ship

NIST AI RMF (Jan 2023) Manage function: monitor, respond, communicate. Ownership lives here, not in the training repo.

Incident Response



A fairness incident is a harm event, not only a metric regression.

Who was affected, which group, which threshold, which override?

Legal notice rules differ: EU AI Act high-risk logs; HK PDPO data-breach guidance; [VERIFY] sector regulator.

Quiz and Discussion

Q1

Why might measuring fairness conflict with PDPO / privacy law?

Q2

PKU-SafeRLHF splits helpfulness and harmlessness. Why is one scalar reward a fairness risk?

Q3

If a RAG index under-covers Cantonese legal sources, which harm is that?

Q4

Name one artifact you would demand in an incident file.

Law and Standards Overview

Instrument	Jurisdiction	Date	Type
NIST AI RMF 1.0	United States	26 Jan 2023	Voluntary framework
EU AI Act	European Union	In force 1 Aug 2024	Binding regulation; staged application
Interim Measures for Generative AI	Mainland China	Effective 15 Aug 2023	Binding administrative measures
PCPD AI / PDPO guidance	Hong Kong SAR	2021 guidance; 2024 Model Framework	Soft law under PDPO

Always pair a rule with jurisdiction and date. EU full application of many duties: 2 Aug 2026 (European Commission).

NIST AI RMF

NIST.AI.100-1, 26 January 2023. Voluntary. Core functions:

GOVERN

Policies, roles, culture, accountability.

MAP

Context, intended use, impacts, third parties.

MEASURE

Metrics, evaluation, including fairness and bias tests.

MANAGE

Prioritize, respond, monitor, communicate residual risk.

Trustworthy AI characteristics include validity, safety, security, accountability, transparency, privacy, fairness / bias management. Generative profile: NIST.AI.600-1, 26 July 2024. RMF 1.0 is being revised [VERIFY current draft].

EU, Mainland China, Hong Kong

EU AI Act

Risk tiers; prohibited practices; high-risk duties.

In force 1 Aug 2024; many duties apply 2 Aug 2026.

Art. 10 data governance for high-risk systems.

Mainland China

CAC + six agencies, 10 Jul 2023. Effective 15 Aug 2023.

Public generative services: filing, security, content, labeling.

Hong Kong

No dedicated AI Act as of this lecture.

PDPO + PCPD: Ethical AI Guidance (2021); Model Personal Data Protection Framework (2024).

Agentic AI privacy guidance, Aug 2026 (PCPD).

Accountability and Explanation

Who is accountable

Developer, deployer, data owner, reviewer.

Write names, not “the model.”

EU: provider vs deployer roles.

What to explain

To a user: why this decision, how to contest.

To an auditor: data, metric, threshold, override.

SHAP $\neq$ accountability if the construct is wrong.

Records

Model card, datasheet, dashboard snapshot, incident log.

SafeLawBench-style legal tests if the product gives legal-adjacent answers.

Participatory Design

Who sits at the table

Affected users, not only data scientists.

They help choose the metric, the harm, the override.

Sloane et al., Participation is not a Design Fix, 2020 —
avoid extractive workshops.

Minimum practice

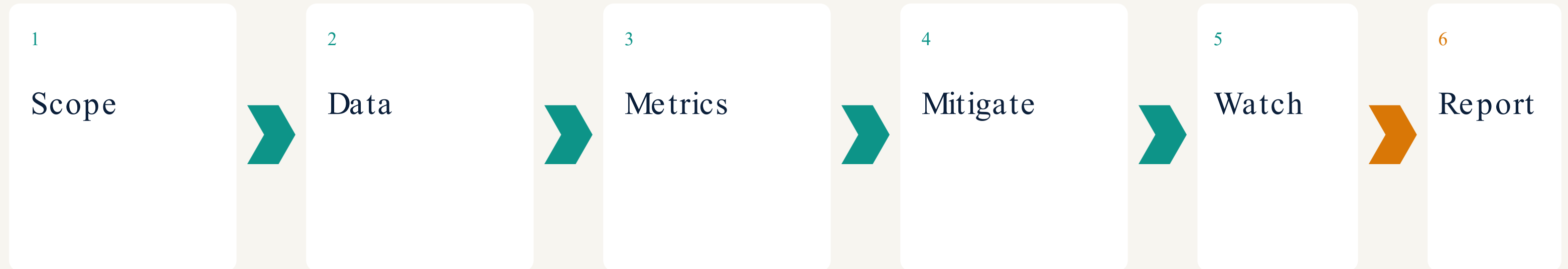
Pay people for their time.

Show the actual interface, not a slide.

Give a written reply on what changed.

If you cannot change the product, do not call it co-design.

Capstone Audit Workflow



Scope: system, population, jurisdiction, decision.

Data: datasheet, missingness, proxies. Metrics: named criteria + CIs + intersections.

Mitigate: pre / in / post / HITL, with residual risk. Watch + report: owner, dashboard, incidents.

Worked Example

Sketch: a Hong Kong public-facing legal Q&A assistant (not a court system).

Scope / data

Users: EN / 繁 / 粵.

Index: ordinances, PCPD, judgments.

Risk: inaccurate legal advice; dialect omission.

Metrics

SafeLawBench-style legal accuracy / refusal.

PrivaCI-Bench-style flow violations.

Exposure of HK vs non-HK sources in RAG.

Controls

Human lawyer review above a risk tier.

Cite retrieved clause; refuse when out of index.

PDPO minimization; no silent logging of client facts.

Group Activity Rubric

Criterion	Pts	What “excellent” looks like
Scope and jurisdiction	15	Named system, population, law, date
Bias localization	20	Lifecycle stage, construct vs label, proxies
Metrics and trade-off	25	Named criteria, impossibility acknowledged, CIs
Mitigation and monitoring	25	Pre/in/post or HITL+ owner + incident path

Remaining 15 pts: sources. No invented statistics. Team of 3–4; 12-minute briefing.

Takeaways

1

Bias, fairness, and harm are different objects. Name all three.

2

Accuracy is not fairness. Build per-group matrices.

3

You cannot have calibration and both balance conditions unless base rates match or Y is perfect.

4

Dropping A does not drop proxies. Intersections hide in averages.

5

Mitigations are staged controls. Residual risk needs an owner.

6

Law is jurisdictional. Pair every claim with a date and an instrument.

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